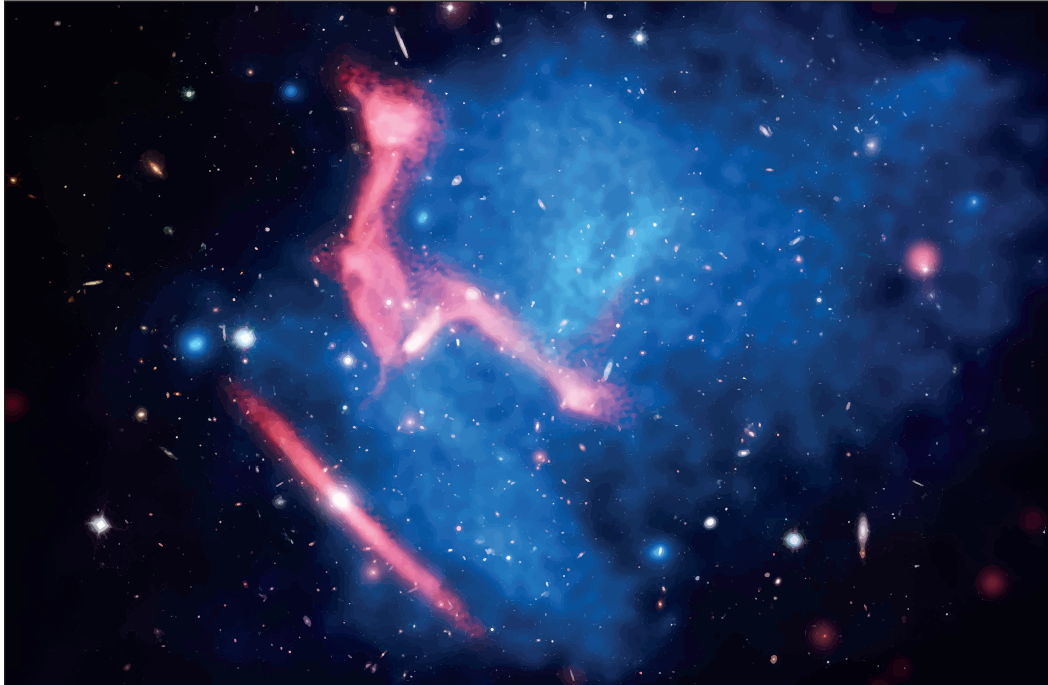


6. Three images have been combined below to show colliding galaxy clusters 5 billion light years from Earth. The background is a Hubble Space Telescope image taken using visible light; the blue area is an X-ray image from the Chandra Space Observatory, and the red area is an image from a radio telescope.



- (a) (i) Compare the use of Earth-based and space-based telescopes to produce images of objects in space. [2]

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- (ii) Explain how the visible light received by the Hubble Space Telescope provides evidence for the origin of the Universe. [3]

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- (b) (i) State how long it takes radio waves to travel from the colliding galaxy clusters to reach the Earth. [1]

time =

- (ii) X-rays have a higher frequency than radio waves. Tick (✓) the box next to the correct statement below. [1]

X-rays take less time to travel the same distance as radio waves.

☐

X-rays take the same time to travel the same distance as radio waves.

☐

X-rays take more time to travel the same distance as radio waves.

☐

- (iii) The X-rays detected by the Chandra Observatory have a frequency range of 3×10^{16} Hz to 3×10^{19} Hz. They travel through space at a speed of 3×10^8 m/s.

Use the equation

$$\text{wave speed} = \text{frequency} \times \text{wavelength}$$

to calculate the maximum wavelength of these X-rays. [3]

maximum wavelength = m

☐

10

END OF PAPER





THE PERIODIC TABLE

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